

Name: _____

Date: _____

ANGLES AND GEOMETRY, (ANGLES IN TRIANGLES AND QUADRILATERALS)

LO: I can find missing angles in triangles and quadrilaterals using angle sum facts.

Angle sums

The angles in a **triangle** always add up to **180°**. The angles in a quadrilateral always add up to 360°.

Types of triangle: Equilateral (all 60°), isosceles (two equal), scalene (all different), right-angled (one 90°).

Key facts

●	Triangle angles sum to 180°	always
●	Quadrilateral angles sum to 360°	always
●	Isosceles: two angles are equal	base angles
●	Triangle angles sum to 360°	that is a quadrilateral

Key idea

Triangle = 180°. Quadrilateral = 360°. Subtract known angles to find the missing one.

$$a + b + c = 180^\circ \text{ (triangle)}$$

Sum of angles in a triangle

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - triangle missing angle

A triangle has angles 50° and 70°. Find the third angle.

Angles sum to 180°

$$50 + 70 = 120$$

$$180 - 120 = 60^\circ$$

$$= 60^\circ$$

Always subtract the sum of known angles from 180°.

Example 2 - isosceles triangle

An isosceles triangle has a top angle of 40°. Find the base angles.

Base angles are equal; call them x

$$40 + x + x = 180 \quad 40 + 2x = 180$$

$$2x = 140 \quad x = 70^\circ$$

$$= 70^\circ \text{ each}$$

In isosceles triangles, the two base angles are always equal.

Example 3 - quadrilateral missing angle

A quadrilateral has angles 85°, 110°, 90° and x. Find x.

Angles sum to 360°

$$85 + 110 + 90 = 285$$

$$x = 360 - 285 = 75^\circ$$

$$x = 75^\circ$$

Quadrilateral angle sum is 360°.

Example 4 - algebraic

A triangle has angles x, 2x and 3x. Find x.

$$x + 2x + 3x = 180$$

$$6x = 180$$

$$x = 30^\circ$$

$$x = 30^\circ$$

Form an equation and solve. Angles are 30°, 60°, 90°.

YOUR TURN – PRACTICE (deliberate practice)

1. Missing angle in a triangle

Find the missing angle.

- a) $60^\circ, 80^\circ, ?$
- b) $45^\circ, 90^\circ, ?$
- c) $30^\circ, 30^\circ, ?$
- d) $55^\circ, 65^\circ, ?$

2. Isosceles triangles

Find the base angles (marked x).

- a) Top angle = 50°
- b) Top angle = 80°
- c) Top angle = 100°
- d) Top angle = 36°

3. Quadrilateral missing angles

Find x in each quadrilateral.

- a) $90^\circ, 90^\circ, 90^\circ, x$
- b) $80^\circ, 100^\circ, 120^\circ, x$
- c) $75^\circ, 105^\circ, 95^\circ, x$
- d) $110^\circ, 70^\circ, 130^\circ, x$

4. Algebraic angles

Form and solve an equation.

- a) Triangle: x, x, x + 60
- b) Triangle: 2x, 3x, 4x
- c) Quadrilateral: x, 2x, 3x, 4x
- d) Triangle: x + 10, 2x, x + 30

5. Mixed problems

Find the missing angle(s).

- a) Right-angled triangle: $90^\circ, 35^\circ, ?$
- b) Equilateral triangle: all angles?
- c) Parallelogram: one angle is 70° ; find all
- d) Kite: $100^\circ, 60^\circ, 100^\circ, ?$
- e) Triangle: $2x - 10, x + 20, x + 30$; find x

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Angles in a triangle sum to 180° ☐

Correct answer: _____

Reason: _____

b) Angles in a quadrilateral sum to 180° ☐

Correct answer: _____

Reason: _____

c) An isosceles triangle has two equal angles ☐

Correct answer: _____

Reason: _____

d) An equilateral triangle has angles of 90° ☐

Correct answer: _____

Reason: _____

e) A right angle is exactly 90° ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) An isosceles triangle has one angle of x° . Explain why x could be the top angle or a base angle. Find all possible triangles if $x = 50^\circ$.

Expression: _____

Simplified: _____

b) A quadrilateral has angles in the ratio **1 : 2 : 3 : 4**. Find each angle. What type of quadrilateral could this be?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

